

COSC2129/1476

Artificial Intelligence



Reasoning under Uncertainty: Uncertainty and Probability

Road Map for Today

- ❑ Automated planning revisited
- ❑ Uncertainty and rational decisions
- ❑ Probabilities
 - Basic concepts
 - Unconditional and conditional probabilities
 - Probability distributions
 - Axioms of probability
 - Inference by enumeration

Automated planning revisited

Planning

- Knowledge representation for planning: PL, FOL state description
- Operator knowledge: Logic representation, PDDL/ STRIPS

Planning algorithms

- Forward and backward state-space search
- Heuristics

Reasoning under Uncertainty



Theoretical reasoning:

-- To decide what to believe.

Practical reasoning:

-- To decide how to act.

“When it is not in our power to determine what is true, we ought to follow what is most probable.” — René Descartes

A Decision Making Scenario

You are considering to buy a used car...



- Is the used car in good condition?
- How much are you willing to pay?
- Should you first get it inspected by a mechanic?
- Should you apply for the car loan?
-

Another Decision Making Scenario



How long does it take to get to the airport?

How long before the flight should you leave home?

30 minutes? “Might be okay”

90 minutes? “Should be fine”

150 minutes? “Yes, that will work”

12 hours? “What! Way too early!!”

Uncertainty

Agent may need to handle uncertainty due to partial observability, non-determinism, or both.

Solution: keeping track of a belief state, i.e., a representation of the set of all possible world states that it might be in, and generating a contingent plan that handles every possible eventuality.

Drawbacks:

- Large and complex belief-state representations.
- A correct contingent plan can grow arbitrarily large.
- Sometimes no plan is guaranteed to achieve the goal, but the agent must have some ways to compare merits of plans that are not guaranteed.

An example – an automatic taxi with the goal of delivering a passenger to the airport on time.

Plan A_{90} – leaving home 90mins before the flight departs and driving at a reasonable speed.

Uncertainty (cont'd)

A logical agent will not be able to conclude with certainty that Plan A_{90} will get the passenger to the airport in time.

It reaches the weak conclusion that Plan A_{90} will get the passenger to the airport in time, as long as:

- the car does not break down
- the car does not run out of petrol
- no accidents happen on the bridge
- the plane does not leave early
- no train hits the car

... ..

The right thing to do, i.e., the **rational decision**, depends on both the relative importance of various goals and the likelihood that, and degree to which, they will be achieved.

A rational decision is to choose the action that is expected to maximize the agent's performance measure (A_{90} vs. A_{120} vs. A_{1440}).

Uncertainty (cont'd)

Diagnosis almost always involves uncertainty.

Logical reasoning breaks down when dealing with uncertainty.

A simple example – diagnosing a dental patient's toothache

-- A possible rule for dental diagnosis using first-order logic

$$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity})$$

-- The above rule is wrong since not all patients with toothaches have cavities, so we change it to:

$$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity}) \vee \text{Disease}(p, \text{GumDisease}) \\ \vee \text{Disease}(p, \text{ImpactedWisdom}) \vee \dots$$

-- The problem is that we could have an unlimited list. How about a causal rule:

$$\forall p \text{ Disease}(p, \text{Cavity}) \Rightarrow \text{Symptom}(p, \text{Toothache})$$

-- The above rule is still incorrect, since not all cavities cause pain.

Solution: making it **logically exhaustive** – augmenting the left-hand side with all the qualifications required.

Uncertainty (cont'd)

Logic fails mainly because of:

- ❑ Laziness

- It is too much work to list all antecedents or consequences needed to ensure an exceptionless rule, and too hard to use such rules.

- ❑ Theoretical ignorance

- Sometimes there is just no complete theory.

- ❑ Practical ignorance

- Even if we know all the rules, we might be uncertain about a particular instance (no enough information has been collected).

The agent's knowledge can at best provide only a degree of belief in the relevant sentences.

Probability

Probability theory deals with degrees of belief.

Logic vs. probability theory

- Same: the world is composed of facts that do or do not hold in particular cases.
- Different: a logical agents believes any fact to be true or false or has no opinion; a probabilistic agent has a numerical degree of belief in any fact between 0 and 1.

Probability summarizes the uncertainty coming from laziness and ignorance.

“We might not know for sure what afflicts a particular patient, but we believe that there is an 80% chance (i.e., a probability of 0.8) that the patient with a toothache has a cavity.” -- This belief could be derived from (1) statistic data, (2) general dental knowledge, (3) a combination of different evidence.

Probability statements are made with respect to a knowledge state not to the real world, and thus they change with new evidence.

Probability measures degrees of belief NOT degrees of truth.

Utility Theory

Suppose I believe the following:

$$P(A_{25} \text{ gets me there on time} \mid \dots) = 0.04$$

$$P(A_{90} \text{ gets me there on time} \mid \dots) = 0.97$$

$$P(A_{120} \text{ gets me there on time} \mid \dots) = 0.99$$

$$P(A_{1440} \text{ gets me there on time} \mid \dots) = 0.999999$$

Which action to choose?

- Depends on preferences for the possible outcomes of these plans.
- An outcome is a completely specified state.

Utility theory is used to represent and reason with preferences, saying that every state has a degree of usefulness (i.e., utility) to an agent, and the agent will prefer states with higher utility.

(Utility refers to **the quality of being usefulness**)

Utility of a state is relevant to an agent.

Decision Theory

Preferences, as expressed by utilities, are combined with probability in the general theory of rational decisions. -- [Decision Theory](#)

Decision Theory = Probability Theory + Utility Theory

Fundamental idea: an agent is rational if and only if it choose the action that yields the highest expected utility, averaged over all the possible outcomes of the action. -- [Principle of maximum expected utility \(MEU\)](#)

Probability Basics

Logical and probabilistic assertions are both about possible worlds.

Logical: which possible worlds are ruled out; Probabilistic: how probable the various worlds are.

Begin with a set Ω -- the sample space (the set of all possible worlds), e.g., 6 possible worlds of rolling a die.

$\omega \in \Omega$ is any possible world, e.g., rolled number is 1.

The possible worlds are **mutually exclusive** and **exhaustive**.

A *fully specified probability model* associates a numerical probability $P(\omega)$ with every possible world $\omega \in \Omega$ subject to

$$0 \leq P(\omega) \leq 1$$

$$\sum_{\omega} P(\omega) = 1$$

Use die rolling as example: $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

Probability Basics (cont'd)

Probabilistic assertions and queries are usually about sets of possible worlds.
E.g., the rolled number of a die is smaller than 4, a double number is rolled, etc.

In probability theory, these sets are called **events**.

In Als, the sets are described by propositions. For each proposition, its corresponding set contains just those possible worlds in which the proposition holds.

The probability of an event A (a subset of Ω) is defined as:

$$P(A) = \sum_{\omega \in A} P(\omega)$$

E.g., $P(\text{DieRoll} < 4) = P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 1/2$

Propositions describes sets of possible worlds in a factored representation, in which each possible world is represented by a set of variable/value pairs.

Random Variables

In probability theory, variables are random variables with their own domains.

A domain is a set of possible values that a random variable can take on.

Boolean random variable: $\{true, false\}$, e.g., *Doubles* = *true*.

The name of a random variable starts with an uppercase letter, and the names of its values are always lowercase.

Proposition of the form $A = true$ ($A = false$) can be abbreviated as a ($\neg a$).

If no ambiguity occurs, a value per se can be used to stand for a proposition that a particular variable has that value, e.g., *sunny* stands for *Weather* = *sunny*.

Domain: Finite vs. Infinite; Discrete vs. Continuous; Ordered vs. Non-ordered

Elementary propositions can be combined using logical connectives.

Syntax

- ❑ Boolean random variables
E.g., *Cavity* (do I have a cavity?)

- ❑ Discrete random variable (finite or infinite)
E.g., *Weather* is one of (*sunny, rain, cloudy, snow*)
Weather = rain is a proposition
*Values must mutually exclusive and be exhaustive

- ❑ Continuous random variables (bounded or unbounded)
E.g., *Temperature = 21.6*
Temperature < 22.0 is also allowed

- ❑ Arbitrary combination of basic propositions via logical connectives
E.g., $(Temperature < 22.0) \wedge (Weather = rain)$

Prior Probability

- Prior or unconditional probabilities of propositions
e.g., $P(\text{Cavity} = \text{true}) = 0.1$ and $P(\text{Weather} = \text{sunny}) = 0.72$
They refer to degrees of belief in propositions in the absence of any other information.
- Probability distribution on a random variable gives the probabilities for all the possible values of this random variable.
 $P(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)
- Joint probability distribution on a set of random variables gives the probabilities of all combinations of the values of those variables.
 $P(\text{Weather}, \text{Cavity})$ is a 4×2 table of probabilities:

<i>Weather =</i>	<i>sunny</i>	<i>rainy</i>	<i>cloudy</i>	<i>snow</i>
<i>Cavity = true</i>	0.144	0.02	0.016	0.02
<i>Cavity = false</i>	0.576	0.08	0.064	0.08

- Every query about a domain can be answered by the joint distribution because every event is a subset of the sample space.

Conditional Probability

- Conditional or posterior probabilities

e.g., $P(\text{cavity} \mid \text{toothache}) = 0.8$, i.e., given that *toothache* is known

- Condition distributions:

$P(X \mid Y)$ gives the values of $P(X = x_i \mid Y = y_j)$ for each possible i, j pair.

E.g., $P(\text{Cavity} \mid \text{Toothache})$ is 2-element vector of 2-element vectors.

- If we know more, e.g., *cavity* is also given, then we have

$$P(\text{cavity} \mid \text{toothache}, \text{cavity}) = 1$$

- New evidence may be irrelevant, allowing simplification, e.g.,

$$P(\text{cavity} \mid \text{toothache}, \text{sunny}) = P(\text{cavity} \mid \text{toothache}) = 0.8$$

Conditional Probability (cont'd)

- Definition of conditional probability:

$$P(a \mid b) = P(a \wedge b) / P(b) \text{ if } P(b) > 0$$

- Product rule gives an alternative formulation:

$$P(a \wedge b) = P(a \mid b) P(b) = P(b \mid a) P(a)$$

- A general version holds for distributions, e.g.,

$$P(\textit{Weather}, \textit{Cavity}) = P(\textit{Weather} \mid \textit{Cavity}) P(\textit{Cavity})$$

(Viewed as a set of 4×2 equations, *not* a matrix multiplication)

- Chain rule is derived by successive application of product rule:

$$\begin{aligned} P(X_1, \dots, X_n) &= P(X_1, \dots, X_{n-1}) P(X_n \mid X_1, \dots, X_{n-1}) \\ &= P(X_1, \dots, X_{n-2}) P(X_{n-1} \mid X_1, \dots, X_{n-2}) P(X_n \mid X_1, \dots, X_{n-1}) \\ &= \dots \\ &= P(X_1) \prod_{i=2}^n P(X_i \mid X_1, \dots, X_{i-1}) \end{aligned}$$

Conditioning vs. Logical Implication

$P(\neg \text{flies} \mid \text{bird})$: proportion of birds that do not fly, which would be low.

$P(\text{bird} \Rightarrow \neg \text{flies}) = P(\neg \text{bird} \vee \neg \text{flies})$: would be dominated by *non-birds* and so would be high!

$P(\text{bird} \Rightarrow \text{flies})$: would also be high, the probability also being dominated by the non-birds.

Axioms of Probability

❑ For any random variable A : $0 \leq P(A) \leq 1$

❑ $P(\text{True}) = 1$

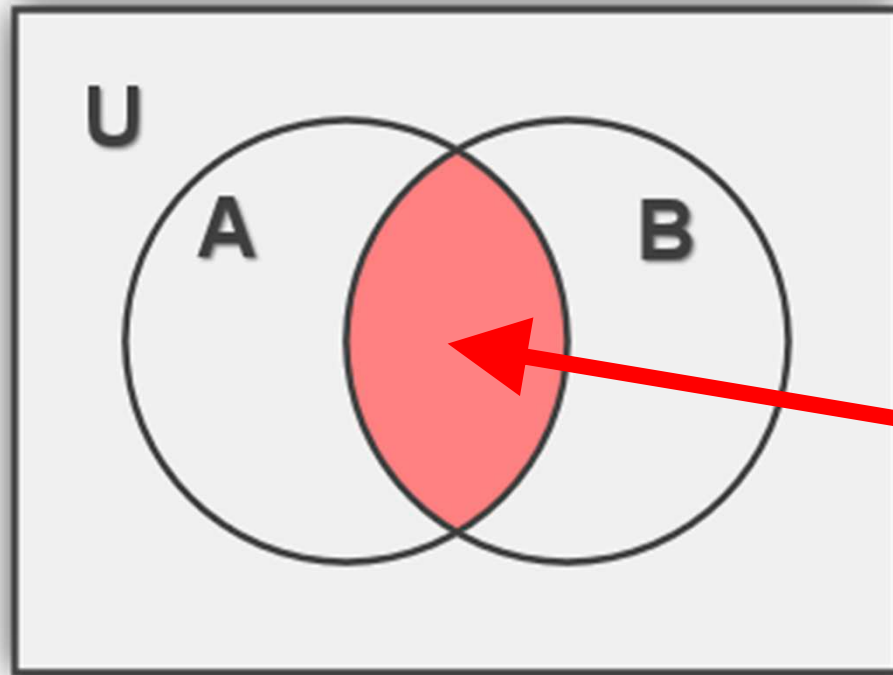
❑ $P(\text{False}) = 0$

❑ $P(A \vee B) = ???$

❑
o $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

Interpreting Axioms

U = sample space, the universe. Its area is $P(U) = 1$ A is an event. Its area is $0 \leq P(A) \leq 1$



$$P(A \vee B) =$$

$$P(A) +$$

$$P(B) -$$

$$P(A \wedge B)$$

Take the Axioms Seriously

- ❑ There have been attempts to use different methodologies for uncertainty.
 - Fuzzy logic, three valued logic, Dempster-Shafer, non-monotonic reasoning,...
- ❑ But if you follow the axioms of probability then no one can take advantage of you 😊.
 - You can always calculate what the best "bet" is.
 - Then you act accordingly.
 - In the long run, it is the "best you can do"...

Theorems from the Axioms

Theorem: $P(\sim A) = 1 - P(A)$

Proof:

$$P(A \vee \sim A) = P(A) + P(\sim A) - P(A \wedge \sim A)$$

$$P(\text{True}) = P(A) + P(\sim A) - P(\text{False})$$

$$1 = P(A) + P(\sim A) - 0$$

$$P(\sim A) = 1 - P(A)$$

PROVED!

Inference by Enumeration

Probabilistic interference: Computing posterior probabilities for query propositions given observed evidence

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

Inference by Enumeration (cont'd)

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

Marginalization or summing out: $P(Y) = \sum_{z \in Z} P(Y, z)$

Conditioning rule: $P(Y) = \sum_{z \in Z} P(Y \mid z) P(z)$

Inference by Enumeration (cont'd)

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

$$P(\text{cavity} \vee \text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 + 0.072 + 0.008 = 0.28$$

Inference by Enumeration (cont'd)

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- Can also compute conditional probabilities:

$$P(\neg \text{cavity} \mid \text{toothache})$$

$$\begin{aligned} &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\ &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 \end{aligned}$$

Inference by Enumeration (cont'd)

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Denominator can be viewed as a normalization constant α

$$P(\text{Cavity} \mid \text{toothache}) = \alpha P(\text{Cavity}, \text{toothache})$$

$$= \alpha [P(\text{Cavity}, \text{toothache}, \text{catch}) + P(\text{Cavity}, \text{toothache}, \neg \text{catch})]$$

$$= \alpha [<P(\text{cavity}, \text{toothache}, \text{catch}), P(\neg \text{cavity}, \text{toothache}, \text{catch})> + \\ <P(\text{cavity}, \text{toothache}, \neg \text{catch}), P(\neg \text{cavity}, \text{toothache}, \neg \text{catch})>]$$

$$= \alpha [<0.108, 0.016> + <0.012, 0.064>]$$

$$= \alpha <0.12, 0.08> = <0.6, 0.4>$$

General idea: compute distribution on query variables by fixing *evidence variables* and summing over *hidden variables*.

Inference by Enumeration (cont'd)

Typically, we are interested in:

the posterior joint distribution of the query variables Y
given specific values e for the evidence variables E

Let the hidden variables be $H = X - Y - E$

Then we can sum out the hidden variables:

$$P(Y \mid E = e) = \alpha P(Y, E = e) = \alpha \sum_h P(Y, E = e, H = h)$$

The terms in the summation are joint entries because Y , E and H together exhaust the set of random variables

Obvious problems:

- Worst-case time complexity $O(d^n)$ where d is the largest arity
- Space complexity $O(d^n)$ to store the joint distribution
- How to find the numbers for $O(d^n)$ entries?

Conclusion

- ❑ Automated planning revisited
- ❑ Uncertainty and rational decisions
- ❑ Probabilities
 - Basic concepts
 - Unconditional and conditional probabilities
 - Probability distributions
 - Axioms of probability
 - Inference by enumeration.